

Validating IBM Fusion installation

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Steps to confirm the success of the installation.

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Health status of individual services [🔗](#)

In the Services page, check the health status of your installed services. The services must be in running or healthy state.

Verify the IBM Fusion installation [🔗](#)

Verify pods from the OpenShift® Container Platform console:

1. Click **View Operator** to view the installed operators in the selected namespace. In this example, it is `ibm-spectrum-fusion-ns` namespace.
2. Go to **Workloads**, **Pods**.
3. Select the namespace where you installed IBM Fusion.
4. Check whether the following pods are running successfully:
 - `isf-application-operator-controller-manager`
 - `isf-cns-operator-controller-manager`
 - `isf-data-protection-operator-controller-manager`
 - `isf-prereq-operator-controller-manager`
 - `isf-serviceability-operator-controller-manager`
 - `isf-ui-operator-controller-manager`
 - `spp-dp-controller-manager`
 - `callhomeclient`
 - `isf-proxy`
 - `isf-ui-dep`
 - `logcollector`
 - `eventmanager`
 - `trapserver`
 - `isf-update-operator-controller-manager`

Alternatively, you can verify the installation by running the oc commands:

```
oc get pods -n ibm-spectrum-fusion-ns
```

A sample result of the oc command is as follows:

```
[root@vm-1527 ~]# oc get pods
NAME                                READY   STATUS    RESTARTS   AGE
callhomeclient-75cb5d95b9-f6nmk     1/1     Running   0           130m
isf-application-operator-controller-manager-7b95c84d6d-fqpbp  2/2     Running   0           137m
isf-cns-operator-controller-manager-6d8754cfd4-lnnlz         2/2     Running   1           137m
isf-data-protection-operator-controller-manager-986bfc476-pzsv6  2/2     Running   1           137m
isf-prereq-operator-controller-manager-597c68796-62cnk       2/2     Running   1           137m
isf-proxy-5bbcbd786b-hhzbj        1/1     Running   0           134m
isf-serviceability-operator-controller-manager-54c555676c-714pq  2/2     Running   0           137m
isf-ui-dep-8486747fff-xjhkw        1/1     Running   0           134m
isf-ui-operator-controller-manager-6b6678dc9c-sqxhk          2/2     Running   1           137m
logcollector-76f696c4bb-lppmg      1/1     Running   0           130m
spp-dp-controller-manager-84b9d984d9-wjwnn                    1/1     Running   1           137m
isf-update-operator-controller-manager-97f666b4f-tc47k        1/1     Running   0
```

Make sure all the pods are up and running.

Verify the IBM Fusion instances as follows:

1. Click **Operators**, **Install operators**. Select the namespace in which IBM Fusion was installed from the Project tab.
2. Select the **IBM Fusion**.
3. Go to **All instance** tab. It lists the IBM Fusion instances that are created. Ensure that the status shows completed for all instances.
The following instances get created and can be validation points for IBM Fusion post installation.

- spectrumfusion CR instance of Kind SpectrumFusion
- data-cataloging-service-definition CR instance of Kind FusionServiceDefinition
- data-foundation-service CR instance of Kind FusionServiceDefinition
- ibm-backup-restore-agent-service CR instance of Kind FusionServiceDefinition
- ibm-backup-restore-service CR instance of Kind FusionServiceDefinition
- Cluster object of kind Cluster, the name of the object can be the DNS entry of the cluster.
For example: apps.sds-auto3.cp.fyre.ibm.com

- CR instance objects of Kind Application with names:

```
ibm-backup-restore
ibm-data-cataloging
ibm-spectrum-fusion-ns
rook-ceph
```

- CR instance objects of Kind BackupStorageLocation with names:

```
in-place-snapshot
isf-dp-inplace-snapshot
```

- Recipe object with name fusion-cr-backup

To open the IBM Fusion user interface in the OpenShift Container Platform web console, click the **Applications** icon in the title bar and select IBM Fusion.

Note: You can use IBM Fusion user interface only when the installation of IBM Fusion is completed successfully.

Verify the Storage or Global data platform installation [↗](#)

- Note:**
- Ensure that the Global data platform service is enabled before validation.
 - IBM Fusion 2.8.0 or higher supports IBM Storage Scale 5.2.0.

Verify whether the IBM Storage Scale installed successfully.

Note: Before you validate, ensure that the IBM Storage Scale is used as the storage provider.

Note: The ibm-spectrum-scale namespace does not have any pods until the remote mount is configured from IBM Fusion UI, that is, the

1. Run the following command to validate the IBM Storage Scale installation.

```
oc describe scalemanager -n ibm-spectrum-fusion-ns
```

2. Check whether the status at the end of CR reflects as completed.
3. If the IBM Storage Scale installation is still in progress, the status continues to show Installing. Wait and ensure that the IBM Storage Scale installation is completed.

Verify the IBM Storage Scale pods as follows:

1. Verify the IBM Storage Scale pods from the OpenShift Container Platform console:
 - a. Go to **Workloads** > **Pods**.
 - b. Select ibm-spectrum-scale, ibm-spectrum-scale-csi, and ibm-spectrum-scale-operator namespaces. It lists all the pods and their status.

Note: The ibm-spectrum-scale namespace does not have any pods until the remote mount is configured from IBM Fusion UI.

2. Verify the IBM Storage Scale deployments and pods from the IBM Fusion CLI.
After you successfully remote mount the filesystem, run the following commands to validate it.

- a. To list all scale pods, run the following command:

```
oc -n ibm-spectrum-scale get pods
```

Example output:

```
./oc get pods -n ibm-spectrum-scale
NAME                                READY   STATUS    RESTARTS   AGE
ibm-spectrum-scale-gui-0            4/4    Running   0           8m30s
ibm-spectrum-scale-gui-1            4/4    Running   0           4m30s
ibm-spectrum-scale-pmcollector-0    2/2    Running   0           8m
ibm-spectrum-scale-pmcollector-1    2/2    Running   0           6m29s
worker0                             2/2    Running   0           8m30s
worker1                             2/2    Running   0           8m30s
worker2                             2/2    Running   0           8m30s
```

The output lists two GUI pods and a daemon pod for each worker node.

- b. Validate pods in ibm-spectrum-scale-dns:

```
oc get pods -n ibm-spectrum-scale-dns
```

Example output:

```
NAME                                READY   STATUS    RESTARTS   AGE
coredns-98mqn                       1/1    Running   0           8m43s
coredns-9ngqc                       1/1    Running   0           8m43s
coredns-h658w                       1/1    Running   0           8m43s
coredns-nw8dj                       1/1    Running   0           8m43s
coredns-pqh5f                       1/1    Running   0           8m43s
coredns-qzgdk                       1/1    Running   0           8m43s
```

- c. To list all the CSI pods, run the following command:

```
oc get pods -n ibm-spectrum-scale-csi
```

Example output:

```
NAME                                READY   STATUS    RESTARTS   AGE
ibm-spectrum-scale-csi-attacher-5d9684696b-9fvdq  1/1    Running   0           3m33s
ibm-spectrum-scale-csi-attacher-5d9684696b-mp8qn  1/1    Running   0           3m33s
ibm-spectrum-scale-csi-f6n17                 3/3    Running   0           3m33s
ibm-spectrum-scale-csi-j22nt                 3/3    Running   0           3m33s
ibm-spectrum-scale-csi-operator-8488bcd9c-vj2bz   1/1    Running   0           102m
ibm-spectrum-scale-csi-provisioner-5d5fbdffbb-qbgx9 1/1    Running   0           3m33s
ibm-spectrum-scale-csi-resizer-7d78944d7b-m74nm  1/1    Running   0           3m33s
ibm-spectrum-scale-csi-snapshotter-7f6dc898f-m5zdn 1/1    Running   0           3m33s
ibm-spectrum-scale-csi-sw9gk                 3/3    Running   0           3m33s
```

- d. To show the IBM Fusion defined scale cluster, run the following command:

```
oc -n ibm-spectrum-fusion-ns get scalecluster -o yaml
```

- e. To check the scale filesystem CR results, run the following command:

```
oc -n ibm-spectrum-scale get filesystem -o yaml
```

- f. To check the scale remote cluster CR result, run the following command:

```
oc -n ibm-spectrum-scale get remoteccluster -o yaml
```

- g. Run the oc command and check whether Daemon.status.clusterID has a value or not.

```
oc -n ibm-spectrum-scale get daemon -oyaml
```

Verify the Backup & Restore installation [↗](#)

Note: Ensure that the Backup & Restore IBM Fusion service is enabled before validation.

Verify that the Backup & Restore IBM Fusion service operators from the OpenShift Container Platform web console:

1. Go to **Operators**, > **Installed operators** from OpenShift Container Platform web console.
2. Select the project as `ibm-backup-restore`.
3. Verify that the following operators show the status as succeeded.
 - Red Hat Integration - AMQ Streams
 - IBM Fusion Backup and Restore Server
 - IBM Fusion Backup and Restore Agent
 - OADP Operator
4. Alternatively, you can verify the status of operators by running the following `oc` command:

Note: Note that more operators appear in the command line output than in the web console.

```
oc get csv -n ibm-backup-restore
```

A sample result of the `oc` command is as follows:

```
# oc get csv -n ibm-backup-restore
NAME                                DISPLAY                                VERSION  REPLACES                PHASE
amqstreams.v2.6.0-2                AMQ Streams                          2.6.0-2  amqstreams.v2.6.0-1    Succeeded
guardian-dm-operator.v2.8.0        IBM Fusion Backup and Restore Data Mover  2.8.0    Succeeded
guardian-dp-operator.v2.8.0        IBM Fusion Backup and Restore Data Protection  2.8.0    Succeeded
guardian-mongo-operator.v2.8.0     IBM Fusion Backup and Restore Mongo      2.8.0    Succeeded
ibm-dataprotectionagent.v2.8.0     IBM Fusion Backup and Restore Agent      2.8.0    Succeeded
ibm-dataprotectionserver.v2.8.0    IBM Fusion Backup and Restore Server     2.8.0    Succeeded
oadp-operator.v1.3.1               OADP Operator                        1.3.1    oadp-operator.v1.3.0  Succeeded
redis-operator.v2.8.0              IBM Fusion Backup and Restore Redis      2.8.0    Succeeded
```

Make sure that the status shows Succeeded.

Verify the Backup & Restore pods from the OpenShift Container Platform console:

1. Go to **Workloads**, > **Pods**.
2. Select the namespace, where you installed IBM Fusion Backup & Restore. In this case, select `ibm-backup-restore` namespace. It lists all the pods. Ensure that all pods are running.
3. Verify whether the following pods are running successfully:

```
amq-streams-cluster-operator
applicationsvc
backup-location-deployment
backup-service
backuppolicy-deployment
guardian-dm-controller-manager
guardian-dp-operator-controller-manager
guardian-kafka-cluster-entity-operator
guardian-kafka-cluster-kafka-0
guardian-kafka-cluster-kafka-1
guardian-kafka-cluster-kafka-2
guardian-kafka-cluster-zookeeper-0
guardian-kafka-cluster-zookeeper-1
guardian-kafka-cluster-zookeeper-2
guardian-minio-0
guardian-mongo-operator-controller-manager
ibm-dataprotectionagent-controller-manager
ibm-dataprotectionserver-catalog-ibm-backup-restore
ibm-dataprotectionserver-controller-manager
ibm-backup-restoreagent-controller-manager
ibm-backup-restoreserver-controller-manager
job-manager
mongodb-0
mongodb-1
mongodb-2
mongodb-ab-0
openshift-adp-controller-manager
redis-master-0
redis-operator-controller-manager
redis-replicas-0
redis-replicas-1
redis-replicas-2
transaction-manager
velero
```

4. Alternatively, you can verify the installation by running the following `oc` command:

```
oc get pods -n ibm-backup-restore
```

A sample result of the `oc` command is as follows:

NAME	READY	STATUS	R
amq-streams-cluster-operator-v2.3.0-1-7d6fb79d84-jdkfh	1/1	Running	0
applicationsvc-55c9b4d6c9-6hdv7	1/1	Running	0
b0f64f9161e0882f278dde2eaa1ea9677f4a230a29180fcf21fc665761hvvxz	0/1	Completed	0
backup-location-deployment-6b565b856c-j4vjc	1/1	Running	0
backup-service-54bf9988f6-47bpv	1/1	Running	0
backuppolicy-deployment-b997cc9bf-dfwnh	1/1	Running	0
bc6176f08ef686ccde24395724b77ea07a586a0bd1fa27ebfd5d704d0dxv9pl	0/1	Completed	0
e349f7c16f02ad6c0c31e41ba2fb1750d5154b58537224d239fe47508872cj5	0/1	Completed	0
f3dd0cbe8cb98614cf163fc5372733148236ea2bdeb5efc6a5d5afe4c085qlk	0/1	Completed	0
ff53c6d827e0fd610a4392c4f9411beb0c785572d2fca1bda57e208650bwv2m	0/1	Completed	0
ffacc5cdaa1e0aa4f3b3c0021f3c87931aa7422f8415303d95457febc8nmzk7	0/1	Completed	0
guardian-dm-controller-manager-64d57bf9ff-28dqj	2/2	Running	0
guardian-dp-operator-controller-manager-6f6d55f6f7-fhndb	2/2	Running	0
guardian-kafka-cluster-entity-operator-b59d699f7-5qxt8	3/3	Running	0
guardian-kafka-cluster-kafka-0	1/1	Running	0
guardian-kafka-cluster-kafka-1	1/1	Running	0
guardian-kafka-cluster-kafka-2	1/1	Running	0
guardian-kafka-cluster-zookeeper-0	1/1	Running	0
guardian-kafka-cluster-zookeeper-1	1/1	Running	0
guardian-kafka-cluster-zookeeper-2	1/1	Running	0
guardian-minio-0	1/1	Running	0
guardian-mongo-operator-controller-manager-6f47776cb4-s6tkm	2/2	Running	0
ibm-dataprotectionagent-controller-manager-54d66f7975-lgdgh	2/2	Running	0
ibm-dataprotectionserver-catalog-ibm-backup-restore-k967t	1/1	Running	0
ibm-dataprotectionserver-controller-manager-749554d89f-q6gmX	2/2	Running	0
job-manager-859484bfc5-fzpt8	1/1	Running	0
mongodb-0	2/2	Running	1
mongodb-1	2/2	Running	1
mongodb-2	2/2	Running	1
mongodb-ab-0	1/1	Running	0
openshift-adp-controller-manager-59fb9f86f4-gdbhb	1/1	Running	0
redis-master-0	1/1	Running	0
redis-operator-controller-manager-647cf89ff7-wl8w2	2/2	Running	0
redis-replicas-0	1/1	Running	0
redis-replicas-1	1/1	Running	0
redis-replicas-2	1/1	Running	0
transaction-manager-5d9b59cf9f-hdzjm	2/2	Running	0
velero-777d65c9b7-455fv	1/1	Running	0

Verify the Backup & Restore IBM Fusion service installation status from the OpenShift Container Platform web console:

- Go to **Operators** > **Installed operators** from OpenShift Container Platform web console.
- Select the namespace as **ibm-backup-restore**.
- Select IBM Fusion Backup & Restore server.
- Click the Data Protection server tab and select Data Protection server.
- Select the YAML tab.
- In the status section, makes sure the following:
 - HealthStatuses Shows the health status of all components listed. It may take 5 minutes or more for all components to show up as healthy.
 - Make sure the install status shows Complete and progressPercentage as 100.
- Alternatively, you can verify the installation status by running the following oc command:

```
oc describe dataprotectionserver dataprotectionserver -n ibm-backup-restore
```

A sample result of the oc command is as follows:

```
Status:
  Health Statuses:
    Service Name:  applicationService
    Status:        Healthy
    Service Name:  backuplocation
    Status:        Healthy
    Service Name:  backuppolicy
    Status:        Healthy
    Service Name:  backupservice
    Status:        Healthy
    Service Name:  jobmanager
    Status:        Healthy
    Service Name:  backupagent
    Status:        Healthy
    Service Name:  mongo
    Status:        Healthy
    Service Name:  redis
    Status:        Healthy
    Service Name:  kafka
    Status:        Healthy
  Install Status:
    Progress Percentage: 100
    Retry On Failure:    false
    Status:              Completed
  Installed Version:    2.5.1
  Upgrade In Progress:  false
  Upgrade Status:
    Retry On Failure:    false
```

- Check the status section at the end of the CR and make sure the following:

- The Install Status section shows the status of the install along with the percentage complete.
- The install status shows the install status as Completed when the installation is successfully completed.
- The health status section lists and shows components health status as healthy.

Note: Individual component health statuses may show Unknown or Degraded for up to five minutes, and show a healthy status when

Verify the Fusion Data Foundation [🔗](#)

Verify the Fusion Data Foundation installation was completed successfully from the OpenShift Container Platform console:

1. Go to **Installed Operators** to check whether the Fusion Data Foundation operator is listed and status is Succeeded.
2. Verify pods from the Red Hat®OpenShift Container Platform web console:
 - a. Go to **Workloads**, **Pods**.
 - b. Select the openshift-storage namespace, where the Fusion Data Foundation is installed.
 - c. Check whether the following pods are running successfully:
 - odf-console
 - odf-operator-controller-manager
3. Alternatively, you can verify the installation by running the OC commands:

```
oc get pod -n openshift-storage
```

A sample result of the OC command is as follows:

NAME	READY	STATUS	RESTARTS	AGE
odf-console-849f64fdd7-rqd97	1/1	Running	0	3m3s
odf-operator-controller-manager-5bd4c85d6b-q8g2f	1/1	Running	0	3m3s

4. Ensure that all the pods are up and running.
5. Verify the Fusion Data Foundation status as follows:
 - a. Click **Operators**, **Install operators**. Select the namespace in which IBM Fusion was installed from the Project tab.
 - b. Select the IBM Fusion.
 - c. Go to **Fusion Service Instance** tab and click **odfmanager** from the list.
 - d. Open **YAML** tab and ensure that `.status.health` is **Healthy**.
6. Alternatively, you can verify the status by running the OC commands:

```
oc get fusionserviceinstances.service.isf.ibm.com -n ibm-spectrum-fusion-ns odfmanager -o jsonpath='{.status.health}{"\n"}'
```

7. Verify the local storage operator pods when backing storage type is local as follows:
 - a. Click **Operators**, **Install operators**. Select the namespace in which IBM Fusion was installed from the Project tab.
 - b. Select the IBM Fusion.
 - c. Go to **Fusion Service Instance** tab and click **odfmanager** from the list.
 - d. Click **YAML** tab and check the backingStorageType.

```
- name: backingStorageType
  provided: true
  value: Local
```

- e. Alternatively, you can verify the backing storage type by running the oc commands.

```
oc get fusionserviceinstances.service.isf.ibm.com -n ibm-spectrum-fusion-ns odfmanager -o jsonpath='{.spec.parameters}{"\n"}'
```

- f. If the backing storage type is local, then verify the local storage operator.
Go to **Installed Operators** to check whether the Local Storage operator is listed and its status is Succeeded.

Note: The green check mark indicates that the local storage operator is installed.

- g. Verify pods from the Red HatOpenShift Container Platform web console:
 - i. Go to **Workloads**, **Pods**.
 - ii. Select the openshift-local-storage namespace, where the local storage operator is installed.
 - iii. Check whether the following pods are running successfully: local-storage-operator-xxxxx
For example: local-storage-operator-75c55cf57d-pfjcp
- h. Ensure that all pods are running.
- i. Alternatively, you can verify the installation by running the oc commands:

```
oc get pod -n openshift-local-storage
```

A sample result of the oc command is as follows:

NAME	READY	STATUS	RESTARTS	AGE
diskmaker-discovery-4pw9r	2/2	Running	0	30m
diskmaker-discovery-b4s58	2/2	Running	0	30m
diskmaker-discovery-cc77p	2/2	Running	0	30m
diskmaker-discovery-cqx6h	2/2	Running	0	30m
diskmaker-discovery-kzxrs	2/2	Running	0	30m
diskmaker-discovery-vnhd1	2/2	Running	0	30m
diskmaker-discovery-vvkpd	2/2	Running	0	30m
local-storage-operator-685586b5d6-67nbc	1/1	Running	0	30m

Verify the Data Cataloging installation [🔗](#)

Verify the IBM Data Cataloging installation was completed successfully from the OpenShift Container Platform web console:

1. Run the following command to verify the installed operators.

```
oc -n ibm-data-cataloging get csv
```

NAME	DISPLAY	VERSION	REPLACES
amqstreams.v2.6.0-2	AMQ Streams	2.6.0-2	amqstreams
db2u-operator.v110509.0.1	IBM Db2	110509.0.1	
ibm-spectrum-discover-operator.v216.0.0-1715012933	IBM Storage Discover	216.0.0-1715012933	

2. Run the following command to verify IBM Data Cataloging status.

```
oc -n ibm-data-cataloging get isd
```

Expected output

NAME	READY	SUMMARY	ERROR	AGE	HEALTHSTATUS
data-cataloging-service-instance	True	Awaiting next reconciliation		9d	Healthy

3. Run the following command to verify that all Persistent Volume Claims are bounded.

```
oc -n ibm-data-cataloging get pvc
```

Expected output

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	
activelogs-c-isd-db2u-0	Bound	pvc-7e98eead-9a38-48b6-a880-16d730120796	150Gi	RWO	
activelogs-c-isd-db2u-1	Bound	pvc-7af71786-1529-4b3c-bbd1-b5d1ee733d33	150Gi	RWO	
c-isd-meta	Bound	pvc-1ba20699-ddb8-4600-bf9f-4d637b14da49	50Gi	RWX	
data-c-isd-db2u-0	Bound	pvc-488540db-c951-4f48-b65c-814bf634ca33	300Gi	RWO	
data-c-isd-db2u-1	Bound	pvc-56a3105a-347e-4b55-b11a-d0a1d351d622	300Gi	RWO	
data-isd-sasl-kafka-0	Bound	pvc-3924c9cc-173d-4244-90fa-a234325dc04b	100Gi	RWO	
data-isd-sasl-zookeeper-0	Bound	pvc-3c66397b-ac03-4ed8-841a-e33d83355a23	64Gi	RWO	
data-isd-ssl-kafka-0	Bound	pvc-ce0ebeaf-d2b6-4e40-8c31-e31d45a208e5	100Gi	RWO	
data-isd-ssl-zookeeper-0	Bound	pvc-cd3fdd20-343b-49ca-ad1a-690f8eec3e60	64Gi	RWO	
isd-backup	Bound	pvc-2e077c8e-e992-4e1a-8b98-d24bfdbb5343	250Gi	RWX	
isd-data	Bound	pvc-b98e124f-1f98-43e5-9bf6-c8758c681192	250Gi	RWX	
tempts-c-isd-db2u-0	Bound	pvc-7fdc9ac5-9158-48ac-bad3-3c24590f1f2b	50Gi	RWO	
tempts-c-isd-db2u-1	Bound	pvc-a2adb668-a362-4f61-8ed3-ae103f9b2cd3	50Gi	RWO	

4. Run the following command to verify Db2u cluster detailed progress.

```
oc -n ibm-data-cataloging get formations.db2u.databases.ibm.com isd -o go-template='{{range .status.components}}{{printf "%s,%s,%s\n" %s %s %s}}
```

Expected output

account	account-ibm-data-cataloging-isd	OK
PersistentVolumeClaim	c-isd-meta	OK
secret	c-isd-sshkeys-db2uhausr	OK

secret	c-isd-sshkeys-db2uadm	OK
secret	c-isd-sshkeys-db2instusr	OK
secret	c-isd-ldappassword	OK
secret	c-isd-ldapblueadminpassword	OK
secret	c-isd-instancepassword	OK
secret	c-isd-db2u-lic	OK
secret	c-isd-certs-wv-rest	OK
secret	c-isd-certs-db2u-api	OK
configmap	c-isd-db2uconfig	OK
configmap	c-isd-db2regconfig	OK
configmap	c-isd-db2dbmconfig	OK
configmap	c-isd-db2dbconfig	OK
service	c-isd-ldap	OK
Deployment	c-isd-ldap	OK
service	c-isd-tools	OK
Deployment	c-isd-tools	OK
configmap	c-isd-db2u-api	OK
service	c-isd-db2u-engn-svc	OK
service	c-isd-db2u-head-engn-svc	OK
Job	c-isd-instdb	OK
service	c-isd-etcd	OK
StatefulSet	c-isd-etcd	OK
service	c-isd-db2u-rest-svc	OK
networkpolicy	c-isd-rest-ext	OK
Deployment	c-isd-rest	OK
service	c-isd-db2u-graph-svc	OK
networkpolicy	c-isd-graph-ext	OK
Deployment	c-isd-graph	OK
service	c-isd-db2u-internal	OK
service	c-isd-db2u	OK
StatefulSet	c-isd-db2u	OK
networkpolicy	c-isd	OK
networkpolicy	c-isd-ext	OK
Job	c-isd-restore-morph	OK

5. Run the following command to verify the Db2u cluster CR status.

```
oc -n ibm-data-cataloging get db2ucluster
```

Expected output

NAME	STATE	MAINTENANCESTATE	AGE
isd	Ready	None	9d

6. Run the following command to verify that the ports 50000 and 50001 are listing.

```
oc exec -n ibm-data-cataloging -it $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --selector name=dashmpp-head-0
```

Expected output

Defaulted container "db2u" out of: db2u, init-labels (init), init-kernel (init)

(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)

Active Internet connections (only servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State	PID/Program name
tcp	0	0	10.254.4.11:60000	0.0.0.0:*	LISTEN	23707/db2sysc 0
tcp	0	0	10.254.4.11:60001	0.0.0.0:*	LISTEN	23712/db2sysc 1
tcp	0	0	0.0.0.0:9443	0.0.0.0:*	LISTEN	-
tcp	0	0	0.0.0.0:50022	0.0.0.0:*	LISTEN	-
tcp	0	0	0.0.0.0:50000	0.0.0.0:*	LISTEN	23707/db2sysc 0
tcp	0	0	0.0.0.0:50001	0.0.0.0:*	LISTEN	23707/db2sysc 0
tcp6	0	0	:::50052	:::*	LISTEN	-
tcp6	0	0	:::50022	:::*	LISTEN	-

7. Run the following command to verify that workload is PUREDATA_OLAP.

```
oc exec -n ibm-data-cataloging -c db2u -it $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --selector name=dashmpp
```


Expected output

DB2_WORKLOAD=PUREDATA_OLAP

8. Run the following command to verify the database encryption configurations.

```
oc -n ibm-data-cataloging exec -c db2u $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --selector name=dashmpp-headless)
```

Encrypted database

= YES

9. Run the following command to verify the Kafka CRs readiness.

```
oc -n ibm-data-cataloging get kafka
```

Expected output

NAME	DESIRED KAFKA REPLICAS	DESIRED ZK REPLICAS	READY	WARNINGS
isd-sasl	1	1	True	
isd-ssl	1	1	True	

10. Run the following command to verify that the IBM® Spectrum Discover pods are running.

```
oc -n ibm-data-cataloging get pod -l component=discover
```

Expected output

NAME	READY	STATUS	RESTARTS	AGE
isd-api-7564475d98-cs6nx	1/1	Running	0	9d
isd-auth-5fcb8bd95f-jvzwp	1/1	Running	0	9d
isd-backup-restore-689fc64f7-pgx22	1/1	Running	0	9d
isd-connmgr-7d4cdb4cc7-xnkzm	1/1	Running	1 (9d ago)	9d
isd-consumer-ceph-le-5f9b5d4676-4fsgg	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-9k9cs	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-bphqp	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-cc2pz	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-chlvk	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-fchp5	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-jlbrw	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-l78cc	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-lpcsn	1/1	Running	0	9d
isd-consumer-ceph-le-5f9b5d4676-n2wqt	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-54qm9	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-6hsq9	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-9p4p5	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-dhlgc	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-hr76h	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-mkvx8	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-tvscz	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-vrrd4	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-w59c2	1/1	Running	0	9d
isd-consumer-cos-le-846f5fd97f-xqnc1	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-4klcc	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-8r6w7	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-fqs5m	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-fw8p	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-h78g5	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-lvw15	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-nhznb	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-q2l8b	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-z6tfj	1/1	Running	0	9d
isd-consumer-cos-scan-79fb979585-zv2zw	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-5mrrg	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-8rphj	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-b5dbv	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-b5lv2	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-d9gbm	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-f5ixf	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-m8nw6	1/1	Running	0	9d

isd-consumer-file-scan-5c7565bf7b-ntfbb	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-ql55x	1/1	Running	0	9d
isd-consumer-file-scan-5c7565bf7b-qvggm	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-6s75x	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-845bx	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-8kmnf	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-gntbt	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-mlnfb	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-p9lpt	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-rckm8	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-vkmpd	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-vmh8x	1/1	Running	0	9d
isd-consumer-protect-scan-67d5ffb65-zm4gf	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-6dw9h	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-9nhsz	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-dnwpw	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-dz2mf	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-h2ffj	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-hdrxg	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-nf779	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-pf6n4	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-q8gqf	1/1	Running	0	9d
isd-consumer-scale-le-5d56449994-xzmvh	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-49vkb	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-54pb8	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-5sr89	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-98lm9	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-d4zlj	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-fc82j	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-mnsn2	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-r2kc6	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-tglisz	1/1	Running	0	9d
isd-consumer-scale-scan-7bdc7957c6-zbhgv	1/1	Running	0	9d
isd-contentsearchagent-5d7bfbd887-r487x	1/1	Running	1 (9d ago)	9d
isd-db-schema-xm5cn	0/1	Completed	0	9d
isd-db2whrest-6c6bbfcb6-9v6v2	1/1	Running	1 (9d ago)	9d
isd-generate-license-6g94v	0/1	Completed	0	9d
isd-importtags-64498dbdd8-5kqpb	1/1	Running	1 (9d ago)	9d
isd-keystone-7dc84b69f5-sw6nn	1/1	Running	0	9d
isd-license-check-27982200-c9gkm	0/1	Completed	0	2d21h
isd-license-check-27983640-ldxv2	0/1	Completed	0	45h
isd-license-check-27985080-rhzjl	0/1	Completed	0	21h
isd-policyengine-598f565867-zg5sn	1/1	Running	1 (9d ago)	9d
isd-producer-ceph-le-76b9885854-557b7	1/1	Running	0	9d
isd-producer-cos-le-85d446c9d8-zvzmg	1/1	Running	0	9d
isd-producer-cos-scan-d584f646d-2t669	1/1	Running	0	9d
isd-producer-file-scan-5586b48fd-zkt7q	1/1	Running	0	9d
isd-producer-protect-scan-7b66bb7c64-wz6qq	1/1	Running	0	9d
isd-producer-scale-le-8d7bc64b4-nvqbl	1/1	Running	0	9d
isd-producer-scale-scan-5df647956-qfl6p	1/1	Running	0	9d
isd-proxy-5f4d7dd8ff-44fwf	1/1	Running	0	9d
isd-reports-c6kmq	0/1	Completed	0	9d
isd-scaleafmdatamover-85cc44b566-644vx	1/1	Running	1 (9d ago)	9d
isd-scaleilmdatamover-6b86dd7b58-j28sn	1/1	Running	1 (9d ago)	9d
isd-sdmonitor-6fffc999b7-s2srd	1/1	Running	0	9d
isd-tikaserver-79548bdb4-ssft5	1/1	Running	0	9d
isd-ui-backend-f6c4b65d7-dwzrs	1/1	Running	0	9d
isd-ui-frontend-9659fc847-vcbm	1/1	Running	0	9d

11. Run the following command to verify the console route availability.

```
oc -n ibm-data-cataloging get route console
```

Expected output

NAME	HOST/PORT	PATH	SERVICES	PORT	TERMINATION	WILDCARD
------	-----------	------	----------	------	-------------	----------

Verify the IBM Fusion CR [🔗](#)

IBM Fusion installation status can also be checked through IBM Fusion CR.

1.  **Important:** The IBM Fusion namespace `ibm-spectrum-fusion-ns` is used in the following steps for your reference. If IBM Fusion is installed in a different namespace, update the namespace in the following steps.

Run the following command and export IBM Fusion namespace as environmental variable.

```
export FUSION_NS="ibm-spectrum-fusion-ns"
```

2. Run the command to validate the IBM Fusion installation status.

```
oc describe SpectrumFusion/spectrumfusion -n $FUSION_NS
```

Where the `SpectrumFusion` is the kind of the CR and `spectrumfusion` is name of the CR.

3. Navigate to OpenShift Container Platform console, go to **Operators**, **Installed operators**.
4. Select the IBM Fusion namespace from the **Project** drop down and select the **Fusion** tab.
It shows the CR kind and name of the CR.
5. This CR show status and version of all IBM Fusion services as well as the IBM Fusion Install status.

```
Spec:
  Global Data Platform:
    Enable: true
  License:
    Accept: true
```

```
Status:
  Global Data Platform Status:
    Service Enabled:      true
    Health:               Healthy
    Install Status:       Completed
    Install Status Message: IBM Spectrum Scale storage installation is succeeded.
    Install Status Message Code: BMYSS0003
    Is CRC Rendered:      false
    Is Deployable:        true
    Is Machine config Rendered: false
    Is supported:         true
    Is Upgrade Failed:    false
    Is Upgrade In Progress: false
    Max Available Version: 5.2.0
    Progress Percentage:  100
    Upgrade Available:    false
    Version:              5.2.0
  Events: <none>
```

Parent topic:

→ [Deploying IBM Fusion](#)